Yash Kishore Maheshwari

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(617) 777-8735 | Boston, MA | Seeking Jan–Aug 2027 Co-op (4–8 months) or Summer 2027 Internship

EDUCATION

Northeastern University, Boston, MA — Expected May 2029
B.S. in Computer Science, John Martinson Honors Program — GPA: **3.99/4.00** | Dean's List
Coursework: Algorithms, Computer Systems, Systems Security, Databases, Cloud Computing, Object-Oriented Design
Certifications: AWS Certified Solutions Architect – Associate (exam Oct 2026), Google Cybersecurity Professional Cert.

EXPERIENCE

Undergraduate Research Assistant — Jul 2026 – Present
Northeastern Khoury Algorithms and Theory Group | *Chain-of-Thought Complexity* | *PEAK Award* — *Boston, MA*
- Implemented **10** transformer constructions in Python and NumPy from mathematical specifications, building a verification suite covering **1.7M+ test cases**, including exhaustive checks over selected input ranges.
- Replaced one-hot positional encoding with spherical codes, cutting **2,004 dimensions to 37**.
- Backed every reported finding with a re-runnable test in the repo, a standard the faculty lead adopted team-wide.

Teaching Assistant, CS 2100 — Jan 2026 – Present
Northeastern University, Khoury College of Computer Sciences — *Boston, MA*
- Led **2** weekly labs and **5** office hours a week for **200+ students** on data structures, pandas, and graph algorithms, debugging live and covering interview-style problems beyond the syllabus.
- Evaluated homework and codewalks across **8** assignments with feedback on structure, test coverage, and design.
- Brought recurring misconceptions from office hours to course staff, changing how those topics were re-taught.

Data Visualization Lead — Sep 2025 – Present
Northeastern Electric Racing — *Boston, MA*
- Engineered an unattended post-race pipeline in Python, pandas, and Matplotlib that **cuts processing time ~40%** and renders a frame-synced telemetry overlay onto race video for driver debriefs.
- Published **nercardata** to PyPI after gathering requirements from **4** engineering subteams, standardizing sensor queries so new members could pull clean telemetry without learning the database schema.

Sales Intern — Aug 2024
Banke International Properties — *Abu Dhabi, UAE*
- Presented market analyses and proposals to high-net-worth clients, translating figures into plain advice.
- Generated **20+** qualified leads through cold outreach, property viewings, and direct prospecting.

PROJECTS

DepWatch: Vulnerability Intelligence Platform | *Python, Airflow, AWS, Gemini API* — Sep 2026 – Present
- Built the Airflow pipeline behind a vulnerability scanner, landing GHSA and OSV data in MinIO and transforming it into a **33,874-record** PostgreSQL corpus matched against a repo's dependency manifest.
- Added a Gemini API layer that checks whether a flagged package is reachable in code, cutting hundreds of matches to the few worth acting on, then writes each fix as a coding-agent-ready prompt.
- Indexed the corpus by description rather than CVE ID, using Qwen3 embeddings and an HNSW index in pgvector.
- Deploying on AWS ECS Fargate, RDS, and MWAA with Terraform.

ImpactTrail | *Next.js, TypeScript, Prisma, PostgreSQL, Gemini API* — Feb – May 2026
- **Best Overall Winner, HackBeanpot 2026**. Led the **5**-person team and ran the pitch.
- Wrote the prompt layer that turns a resident's complaint into a volunteer quest, feeding the Gemini API every live quest and validating each response against a Zod schema built from the Prisma enums.
- Modeled volunteer-to-quest matching as a **min-cost flow** capped per volunteer and per quest, over a **13**-table schema.
- Set up GitHub Actions CI with type-check, build, and automatic rollback on a failed deploy.

SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL
Frameworks & APIs: React, Next.js, Tailwind CSS, Node.js, FastAPI, REST APIs, pytest, Puppeteer
Data & AI: PostgreSQL, Prisma, Supabase, pgvector, MinIO, Airflow, pandas, NumPy, Matplotlib, Gemini API, RAG
Cloud & Security: AWS (ECS, RDS, MWAA), Docker, Terraform, GitHub Actions, Git, Linux, MQTT, CVE/CVSS, SBOM